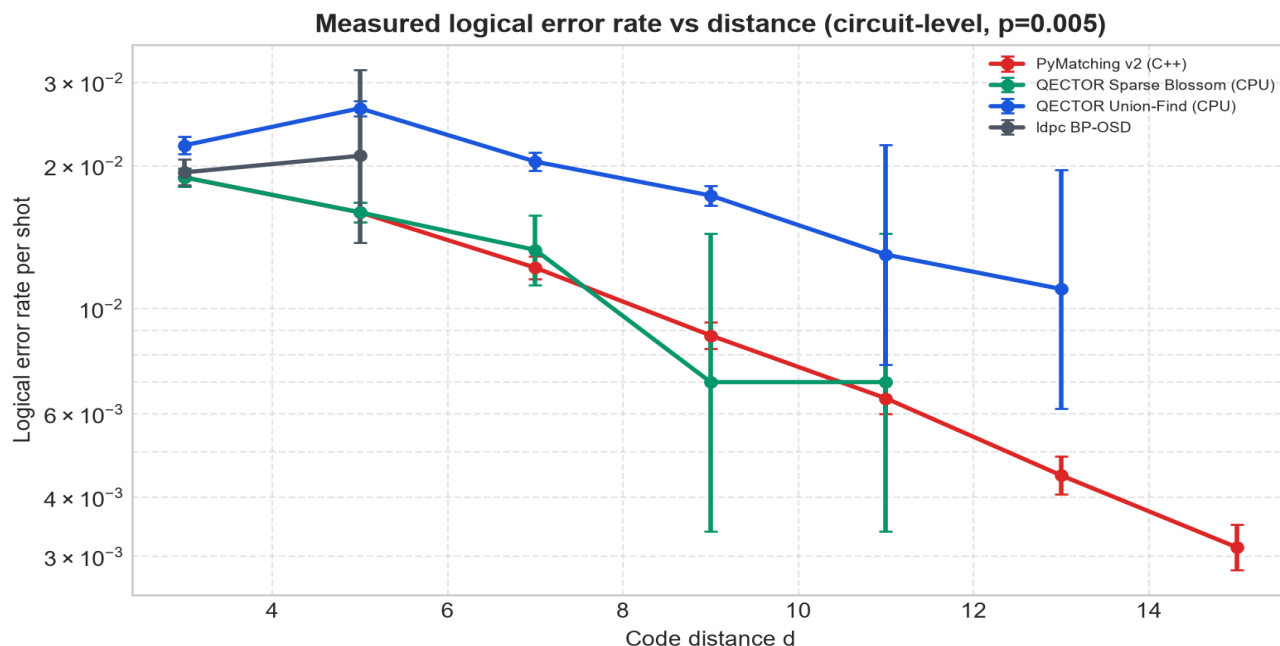
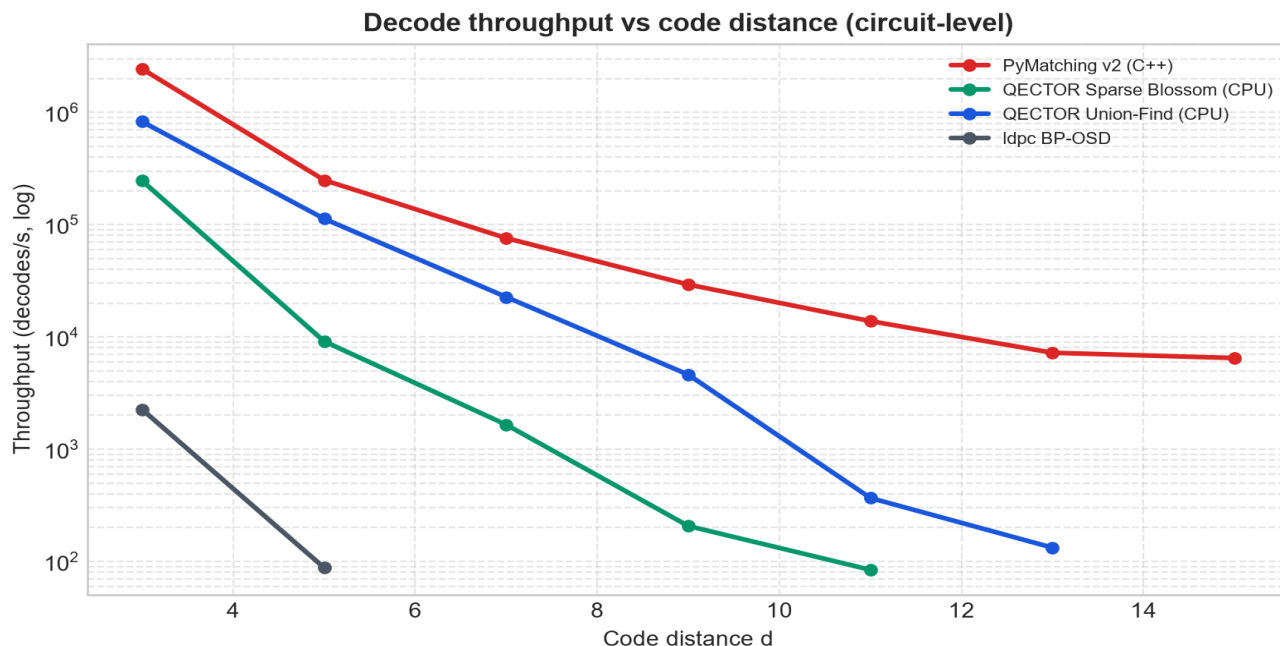


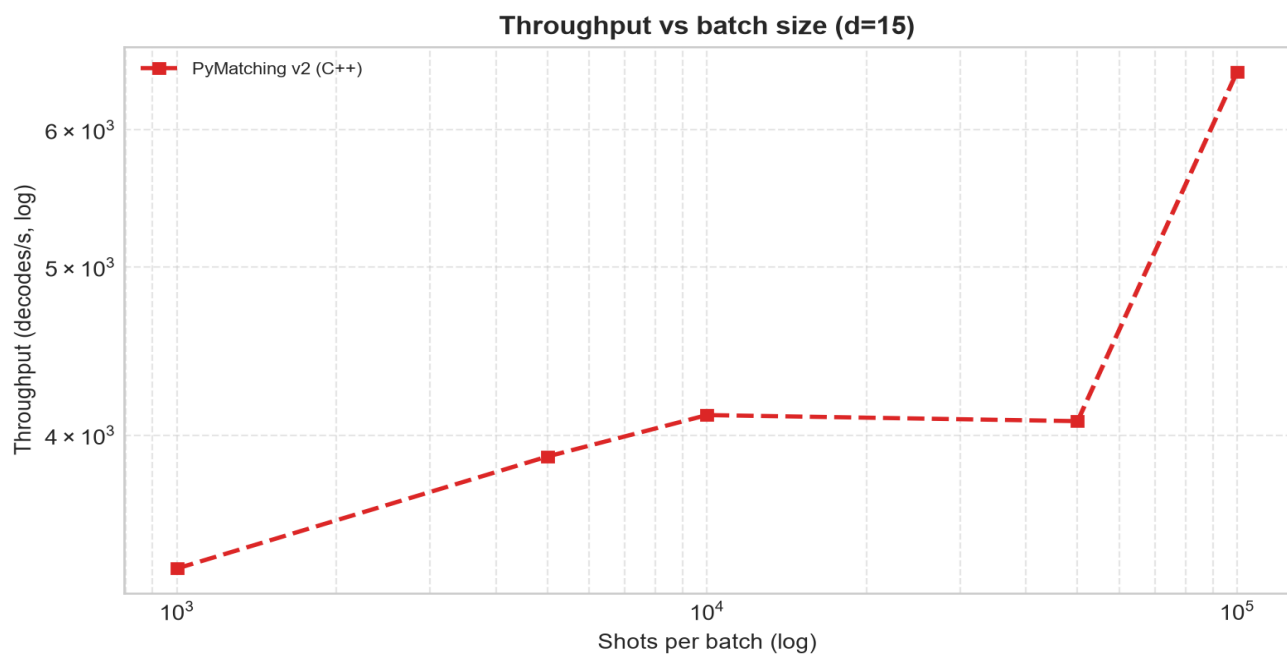
QECTOR v0.7.0 — circuit-level decoder comparison

Measured with `ler.estimate_ler_circuit_level`: identical circuit, DEM, samples and `decode_batch` resolver for every decoder, scored against the circuit's logical observables. Validated by `ler.assert_comparable`. No extrapolated cells.

Noise model: circuit-level, $p=0.005$ | Commit: b0f5456470 | Generated: 2026-07-31T23:03:23Z

Platform: Windows-10-10.0.26100-SP0 | Python: 3.11.9





Measured results (77 cells)

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
PyMatching v2 (C++)	3	1,000	13	0.01300	[0.0076, 0.0221]	1,949,318	1.00x
QECTOR Sparse Blossom (CPU)	3	1,000	13	0.01300	[0.0076, 0.0221]	154,269	0.08x
QECTOR Union-Find (CPU)	3	1,000	16	0.01600	[0.0099, 0.0258]	222,321	0.11x
ldpc BP-OSD	3	1,000	12	0.01200	[0.0069, 0.0209]	2,267	0.001x
PyMatching v2 (C++)	3	5,000	84	0.01680	[0.0136, 0.0208]	1,865,811	1.00x
QECTOR Sparse Blossom (CPU)	3	5,000	84	0.01680	[0.0136, 0.0208]	236,337	0.13x
QECTOR Union-Find (CPU)	3	5,000	105	0.02100	[0.0174, 0.0254]	481,024	0.26x
ldpc BP-OSD	3	5,000	87	0.01740	[0.0141, 0.0214]	2,286	0.001x
PyMatching v2 (C++)	3	10,000	202	0.02020	[0.0176, 0.0231]	1,848,941	1.00x
QECTOR Sparse Blossom (CPU)	3	10,000	202	0.02020	[0.0176, 0.0231]	248,616	0.13x
QECTOR Union-Find (CPU)	3	10,000	233	0.02330	[0.0205, 0.0264]	571,932	0.31x
ldpc BP-OSD	3	10,000	199	0.01990	[0.0173, 0.0228]	2,330	0.001x
PyMatching v2 (C++)	3	50,000	960	0.01920	[0.0180, 0.0204]	2,234,837	1.00x
QECTOR Sparse Blossom (CPU)	3	50,000	960	0.01920	[0.0180, 0.0204]	250,396	0.11x
QECTOR Union-Find (CPU)	3	50,000	1128	0.02256	[0.0213, 0.0239]	812,806	0.36x
ldpc BP-OSD	3	50,000	969	0.01938	[0.0182, 0.0206]	2,259	0.001x
PyMatching v2 (C++)	3	100,000	1891	0.01891	[0.0181, 0.0198]	2,437,651	1.00x
QECTOR Sparse Blossom (CPU)	3	100,000	1891	0.01891	[0.0181, 0.0198]	245,629	0.10x
QECTOR Union-Find (CPU)	3	100,000	2210	0.02210	[0.0212, 0.0230]	825,726	0.34x
PyMatching v2 (C++)	5	1,000	19	0.01900	[0.0122, 0.0295]	266,411	1.00x
QECTOR Sparse Blossom (CPU)	5	1,000	19	0.01900	[0.0122, 0.0295]	8,690	0.03x
QECTOR Union-Find (CPU)	5	1,000	26	0.02600	[0.0178, 0.0378]	79,888	0.30x
ldpc BP-OSD	5	1,000	21	0.02100	[0.0138, 0.0319]	88	0x
PyMatching v2 (C++)	5	5,000	73	0.01460	[0.0116, 0.0183]	286,110	1.00x
QECTOR Sparse Blossom (CPU)	5	5,000	73	0.01460	[0.0116, 0.0183]	9,018	0.03x
QECTOR Union-Find (CPU)	5	5,000	148	0.02960	[0.0253, 0.0347]	91,363	0.32x
PyMatching v2 (C++)	5	10,000	166	0.01660	[0.0143, 0.0193]	234,845	1.00x
QECTOR Sparse Blossom (CPU)	5	10,000	166	0.01660	[0.0143, 0.0193]	9,204	0.04x
QECTOR Union-Find (CPU)	5	10,000	268	0.02680	[0.0238, 0.0302]	97,208	0.41x
PyMatching v2 (C++)	5	50,000	809	0.01618	[0.0151, 0.0173]	225,455	1.00x
QECTOR Sparse Blossom (CPU)	5	50,000	810	0.01620	[0.0151, 0.0173]	8,990	0.04x
QECTOR Union-Find (CPU)	5	50,000	1356	0.02712	[0.0257, 0.0286]	113,116	0.50x
PyMatching v2 (C++)	5	100,000	1596	0.01596	[0.0152, 0.0168]	249,065	1.00x
QECTOR Sparse Blossom (CPU)	5	100,000	1596	0.01596	[0.0152, 0.0168]	9,121	0.04x
QECTOR Union-Find (CPU)	5	100,000	2645	0.02645	[0.0255, 0.0275]	112,797	0.45x
PyMatching v2 (C++)	7	1,000	11	0.01100	[0.0062, 0.0196]	54,523	1.00x
QECTOR Sparse Blossom (CPU)	7	1,000	11	0.01100	[0.0062, 0.0196]	1,631	0.03x
QECTOR Union-Find (CPU)	7	1,000	18	0.01800	[0.0114, 0.0283]	9,142	0.17x
PyMatching v2 (C++)	7	5,000	74	0.01480	[0.0118, 0.0185]	62,572	1.00x
QECTOR Sparse Blossom (CPU)	7	5,000	75	0.01500	[0.0120, 0.0188]	1,565	0.03x
QECTOR Union-Find (CPU)	7	5,000	107	0.02140	[0.0177, 0.0258]	20,731	0.33x
PyMatching v2 (C++)	7	10,000	130	0.01300	[0.0110, 0.0154]	89,104	1.00x
QECTOR Sparse Blossom (CPU)	7	10,000	133	0.01330	[0.0112, 0.0157]	1,651	0.02x
QECTOR Union-Find (CPU)	7	10,000	209	0.02090	[0.0183, 0.0239]	22,427	0.25x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
PyMatching v2 (C++)	7	50,000	619	0.01238	[0.0114, 0.0134]	69,244	1.00x
QECTOR Union-Find (CPU)	7	50,000	1055	0.02110	[0.0199, 0.0224]	21,552	0.31x
PyMatching v2 (C++)	7	100,000	1220	0.01220	[0.0115, 0.0129]	75,767	1.00x
QECTOR Union-Find (CPU)	7	100,000	2042	0.02042	[0.0196, 0.0213]	22,643	0.30x
PyMatching v2 (C++)	9	1,000	7	0.00700	[0.0034, 0.0144]	16,122	1.00x
QECTOR Sparse Blossom (CPU)	9	1,000	7	0.00700	[0.0034, 0.0144]	207	0.01x
QECTOR Union-Find (CPU)	9	1,000	14	0.01400	[0.0084, 0.0234]	1,404	0.09x
PyMatching v2 (C++)	9	5,000	36	0.00720	[0.0052, 0.0100]	28,259	1.00x
QECTOR Union-Find (CPU)	9	5,000	85	0.01700	[0.0138, 0.0210]	4,692	0.17x
PyMatching v2 (C++)	9	10,000	78	0.00780	[0.0063, 0.0097]	27,630	1.00x
QECTOR Union-Find (CPU)	9	10,000	156	0.01560	[0.0134, 0.0182]	3,812	0.14x
PyMatching v2 (C++)	9	50,000	450	0.00900	[0.0082, 0.0099]	29,152	1.00x
QECTOR Union-Find (CPU)	9	50,000	827	0.01654	[0.0155, 0.0177]	4,599	0.16x
PyMatching v2 (C++)	9	100,000	878	0.00878	[0.0082, 0.0094]	29,244	1.00x
QECTOR Union-Find (CPU)	9	100,000	1732	0.01732	[0.0165, 0.0181]	4,606	0.16x
PyMatching v2 (C++)	11	1,000	7	0.00700	[0.0034, 0.0144]	11,490	1.00x
QECTOR Sparse Blossom (CPU)	11	1,000	7	0.00700	[0.0034, 0.0144]	84	0.007x
QECTOR Union-Find (CPU)	11	1,000	13	0.01300	[0.0076, 0.0221]	367	0.03x
PyMatching v2 (C++)	11	5,000	24	0.00480	[0.0032, 0.0071]	12,517	1.00x
PyMatching v2 (C++)	11	10,000	77	0.00770	[0.0062, 0.0096]	13,693	1.00x
PyMatching v2 (C++)	11	50,000	334	0.00668	[0.0060, 0.0074]	14,040	1.00x
PyMatching v2 (C++)	11	100,000	647	0.00647	[0.0060, 0.0070]	13,816	1.00x
PyMatching v2 (C++)	13	1,000	3	0.00300	[0.0010, 0.0088]	7,384	1.00x
QECTOR Union-Find (CPU)	13	1,000	11	0.01100	[0.0062, 0.0196]	133	0.02x
PyMatching v2 (C++)	13	5,000	18	0.00360	[0.0023, 0.0057]	7,576	1.00x
PyMatching v2 (C++)	13	10,000	37	0.00370	[0.0027, 0.0051]	6,933	1.00x
PyMatching v2 (C++)	13	50,000	221	0.00442	[0.0039, 0.0050]	7,243	1.00x
PyMatching v2 (C++)	13	100,000	445	0.00445	[0.0041, 0.0049]	7,218	1.00x
PyMatching v2 (C++)	15	1,000	2	0.00200	[0.0005, 0.0073]	3,351	1.00x
PyMatching v2 (C++)	15	5,000	12	0.00240	[0.0014, 0.0042]	3,888	1.00x
PyMatching v2 (C++)	15	10,000	44	0.00440	[0.0033, 0.0059]	4,108	1.00x
PyMatching v2 (C++)	15	50,000	171	0.00342	[0.0029, 0.0040]	4,075	1.00x
PyMatching v2 (C++)	15	100,000	314	0.00314	[0.0028, 0.0035]	6,480	1.00x

Not measured (63 cells)

Listed so the gaps are explicit. These cells were never run and carry no numbers; nothing here was extrapolated or synthesised.

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	3	100,000	over per-cell decode budget	41s
ldpc BP-OSD	5	5,000	over per-cell decode budget	61s
ldpc BP-OSD	5	10,000	over per-cell decode budget	122s
ldpc BP-OSD	5	50,000	over per-cell decode budget	609s
ldpc BP-OSD	5	100,000	over per-cell decode budget	1,217s
QECTOR Sparse Blossom (CPU)	7	50,000	over per-cell decode budget	36s
QECTOR Sparse Blossom (CPU)	7	100,000	over per-cell decode budget	71s
ldpc BP-OSD	7	1,000	over per-cell decode budget	53s
ldpc BP-OSD	7	5,000	over per-cell decode budget	263s
ldpc BP-OSD	7	10,000	over per-cell decode budget	526s
ldpc BP-OSD	7	50,000	over per-cell decode budget	2,629s
ldpc BP-OSD	7	100,000	over per-cell decode budget	5,258s
QECTOR Sparse Blossom (CPU)	9	5,000	over per-cell decode budget	26s
QECTOR Sparse Blossom (CPU)	9	10,000	over per-cell decode budget	53s
QECTOR Sparse Blossom (CPU)	9	50,000	over per-cell decode budget	263s
QECTOR Sparse Blossom (CPU)	9	100,000	over per-cell decode budget	525s
ldpc BP-OSD	9	1,000	over per-cell decode budget	119s
ldpc BP-OSD	9	5,000	over per-cell decode budget	597s
ldpc BP-OSD	9	10,000	over per-cell decode budget	1,193s
ldpc BP-OSD	9	50,000	over per-cell decode budget	5,967s
ldpc BP-OSD	9	100,000	over per-cell decode budget	11,935s
QECTOR Sparse Blossom (CPU)	11	5,000	over per-cell decode budget	61s
QECTOR Sparse Blossom (CPU)	11	10,000	over per-cell decode budget	122s
QECTOR Sparse Blossom (CPU)	11	50,000	over per-cell decode budget	609s
QECTOR Sparse Blossom (CPU)	11	100,000	over per-cell decode budget	1,219s
QECTOR Union-Find (CPU)	11	5,000	over per-cell decode budget	29s
QECTOR Union-Find (CPU)	11	10,000	over per-cell decode budget	58s
QECTOR Union-Find (CPU)	11	50,000	over per-cell decode budget	291s
QECTOR Union-Find (CPU)	11	100,000	over per-cell decode budget	581s
ldpc BP-OSD	11	1,000	over per-cell decode budget	271s
ldpc BP-OSD	11	5,000	over per-cell decode budget	1,353s
ldpc BP-OSD	11	10,000	over per-cell decode budget	2,706s
ldpc BP-OSD	11	50,000	over per-cell decode budget	13,530s
ldpc BP-OSD	11	100,000	over per-cell decode budget	27,059s
QECTOR Sparse Blossom (CPU)	13	1,000	over per-cell decode budget	39s
QECTOR Sparse Blossom (CPU)	13	5,000	over per-cell decode budget	194s
QECTOR Sparse Blossom (CPU)	13	10,000	over per-cell decode budget	387s
QECTOR Sparse Blossom (CPU)	13	50,000	over per-cell decode budget	1,937s
QECTOR Sparse Blossom (CPU)	13	100,000	over per-cell decode budget	3,875s
QECTOR Union-Find (CPU)	13	5,000	over per-cell decode budget	75s
QECTOR Union-Find (CPU)	13	10,000	over per-cell decode budget	149s
QECTOR Union-Find (CPU)	13	50,000	over per-cell decode budget	746s
QECTOR Union-Find (CPU)	13	100,000	over per-cell decode budget	1,492s

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	13	1,000	over per-cell decode budget	452s
ldpc BP-OSD	13	5,000	over per-cell decode budget	2,258s
ldpc BP-OSD	13	10,000	over per-cell decode budget	4,516s
ldpc BP-OSD	13	50,000	over per-cell decode budget	22,578s
ldpc BP-OSD	13	100,000	over per-cell decode budget	45,156s
QECTOR Sparse Blossom (CPU)	15	1,000	over per-cell decode budget	95s
QECTOR Sparse Blossom (CPU)	15	5,000	over per-cell decode budget	474s
QECTOR Sparse Blossom (CPU)	15	10,000	over per-cell decode budget	949s
QECTOR Sparse Blossom (CPU)	15	50,000	over per-cell decode budget	4,745s
QECTOR Sparse Blossom (CPU)	15	100,000	over per-cell decode budget	9,489s
QECTOR Union-Find (CPU)	15	1,000	over per-cell decode budget	43s
QECTOR Union-Find (CPU)	15	5,000	over per-cell decode budget	213s
QECTOR Union-Find (CPU)	15	10,000	over per-cell decode budget	426s
QECTOR Union-Find (CPU)	15	50,000	over per-cell decode budget	2,132s
QECTOR Union-Find (CPU)	15	100,000	over per-cell decode budget	4,264s
ldpc BP-OSD	15	1,000	over per-cell decode budget	729s
ldpc BP-OSD	15	5,000	over per-cell decode budget	3,647s
ldpc BP-OSD	15	10,000	over per-cell decode budget	7,294s
ldpc BP-OSD	15	50,000	over per-cell decode budget	36,468s
ldpc BP-OSD	15	100,000	over per-cell decode budget	72,937s